

YEAR 7 • UNIT 1 • EXTRA HOMEWORK

Bring Down the Next Digit

Two-digit multiplication • and long division, written out

Name: _____

Class: _____

Date given: _____

Date due: _____

Why this sheet exists

Every method in Unit 1 finishes in one of **two** operations. You find a factor by **dividing**. You find the HCF by dividing twice. You check a square past 12×12 by **multiplying**, and a cube by multiplying twice.

Last time you divided by counting up in multiples. That works until the number gets big. This sheet is the **written method** — the one that does not fall over on 4 digits.

No calculator, and show your working. Working you can see is working you can check.

Part 1 • The facts both methods run on

E1 Fill in the missing number

Every long multiplication and every long division is these facts, one after another. Aim to know them without counting.

(a) $7 \times 6 =$ _____ (e) $9 \times 8 =$ _____ (i) $9 \times 7 =$ _____ (m) $12 \times 7 =$ _____

(b) $8 \times 7 =$ _____ (f) $7 \times 7 =$ _____ (j) $6 \times 6 =$ _____ (n) $11 \times 8 =$ _____

(c) $9 \times 6 =$ _____ (g) $4 \times 9 =$ _____ (k) $9 \times 9 =$ _____ (o) $12 \times 9 =$ _____

(d) $6 \times 8 =$ _____ (h) $8 \times 8 =$ _____ (l) $8 \times 4 =$ _____ (p) $7 \times 4 =$ _____

E2 Multiply by a ten

The second row is always a ten

To multiply by **20**: multiply by **2**, then **put a zero on the end**. $34 \times 2 = 68$, so $34 \times 20 = 680$. That is the whole trick, and it is what the second row of a long multiplication is.

(a) $26 \times 10 =$ _____ (c) $47 \times 20 =$ _____ (e) $18 \times 60 =$ _____

(b) $26 \times 30 =$ _____ (d) $52 \times 40 =$ _____ (f) $73 \times 50 =$ _____

E3 Count up in a two-digit number**Do this before you divide**

Before dividing by 23, write out the first few multiples of 23. Then the division is only **reading your own list**. This is two minutes that saves ten.

×	1	2	3	4	5	6
23						
16						
24						
37						

Part 2 · Two rows, then add**The method, once, in full**

$$\begin{array}{r}
 58 \\
 \times 34 \\
 \hline
 232 \\
 1740 \\
 \hline
 1972
 \end{array}$$

Row 1: $58 \times 4 = 232$.**Row 2:** $58 \times 30 = 1740$ – multiply by 3, put the zero on.**Add them:** $232 + 1740 = 1972$.

One row means you forgot the tens. Your answer comes out about ten times too small. **Two rows, every time.**

E4 Warm up: two digits by one digit

53×7	68×6	87×9	76×8
94×4	47×8	65×7	89×6
58×9	73×6	96×7	84×5

E5 Two digits by two digits**Two rows, then add.** Show both rows every time.

32×14	45×23	28×36
57×42	63×27	86×35
74×58	49×61	95×24
38×47	84×73	66×52

E6 Three digits by two digits**Nothing new here**

Three digits by two is still **two rows**. Row 1 is the whole number times the units. Row 2 is the whole number times the tens. It is only longer, not harder.

124×32	236×18	407×25
318×46	529×37	324×18

Look back. The last one, 324×18 , is the second half of a cube root: $18 \times 18 = 324$, and then 324×18 . So $\sqrt[3]{\quad} = 18$.

Part 3 • Long division: bring down the next digit

Word help – the words in a division

divisor	the number you are dividing by . In $966 \div 23$ it is 23.
goes into	"23 goes into 96 four times" means $96 \div 23 = 4$, with some left over.
bring down	take the next digit of the big number and write it beside what is left.
remainder	what is left at the very end. We write $75 \div 8 = 9 \text{ r } 3$.

Divide, multiply, subtract, bring down – then start again

$$\begin{array}{r}
 42 \\
 23 \overline{) 966} \\
 \underline{92} \\
 46 \\
 \underline{46} \\
 0
 \end{array}$$

Divide: 23 goes into 96 **four** times. Write the 4 on top, above the 6.

Multiply: $4 \times 23 = 92$. Write it underneath.

Subtract: $96 - 92 = 4$.

Bring down the next digit, the 6, to make **46**.

Now start again: 23 goes into 46 **twice**, $2 \times 23 = 46$, $46 - 46 = 0$. Nothing is left, so $966 \div 23 = 42$.

E7 Warm up: one-digit divisors, no remainder

- | | |
|--------------------------|---------------------------|
| (a) $252 \div 6 =$ _____ | (e) $1344 \div 4 =$ _____ |
| (b) $469 \div 7 =$ _____ | (f) $2016 \div 6 =$ _____ |
| (c) $584 \div 8 =$ _____ | (g) $3528 \div 7 =$ _____ |
| (d) $765 \div 9 =$ _____ | (h) $5832 \div 8 =$ _____ |

E8 One-digit divisors, with a remainder

Write each answer as a whole number and a remainder, like this: $75 \div 8 = 9 \text{ r } 3$.

- | | |
|--------------------------|---------------------------|
| (a) $437 \div 6 =$ _____ | (d) $1000 \div 7 =$ _____ |
| (b) $529 \div 7 =$ _____ | (e) $2345 \div 8 =$ _____ |
| (c) $815 \div 9 =$ _____ | (f) $4567 \div 9 =$ _____ |

E9 Two-digit divisors – set it out properly

These all come out exactly. Show the full method in each box.

Write the list first

Before you start, write the first six multiples of the divisor in the corner of the box. Then every “how many times does it go in” is **reading**, not guessing.

$768 \div 16$

$851 \div 23$

$1224 \div 24$

$2072 \div 37$

$4212 \div 18$

$3375 \div 15$

$1088 \div 32$

$1222 \div 47$

$986 \div 29$

$2408 \div 43$

$1872 \div 26$

$2940 \div 35$

E10 Two-digit divisors, with a remainder

The remainder is always **smaller than the divisor**. Left with 30 after dividing by 24? Then 24 goes in **one more time** — go back.

$1000 \div 24$	$875 \div 32$
$2500 \div 36$	$1500 \div 45$
$3000 \div 28$	$1800 \div 26$

E11 Find the missing number

Every multiplication is two divisions

If $24 \times 51 = 1224$, then $1224 \div 24 = 51$ **and** $1224 \div 51 = 24$. Look at **which** number is missing before you decide what to do: if the **big** number is missing, you **multiply**.

(a) $24 \times \underline{\hspace{2cm}} = 1224$

(i) $43 \times \underline{\hspace{2cm}} = 2408$

(b) $\underline{\hspace{2cm}} \times 16 = 768$

(j) $\underline{\hspace{2cm}} \times 26 = 1872$

(c) $2072 \div \underline{\hspace{2cm}} = 56$

(k) $2940 \div \underline{\hspace{2cm}} = 84$

(d) $\underline{\hspace{2cm}} \div 18 = 234$

(l) $\underline{\hspace{2cm}} \div 29 = 34$

(e) $37 \times \underline{\hspace{2cm}} = 851$

(m) $47 \times \underline{\hspace{2cm}} = 1222$

(f) $\underline{\hspace{2cm}} \times 18 = 324$

(n) $\underline{\hspace{2cm}} \times 32 = 1088$

(g) $3375 \div \underline{\hspace{2cm}} = 225$

(o) $4212 \div \underline{\hspace{2cm}} = 234$

(h) $\underline{\hspace{2cm}} \div 8 = 729$

(p) $\underline{\hspace{2cm}} \div 15 = 225$

E12 Mixed drill — no boxes, no help

Both operations, mixed up. Work down the columns as fast as you can be accurate. **Two pairs of these have the same answer — find both pairs.**

- | | | |
|----------------------------|----------------------------|----------------------------|
| 1. $480 \div 6 =$ _____ | 11. $936 \div 12 =$ _____ | 21. $999 \div 3 =$ _____ |
| 2. $24 \times 30 =$ _____ | 12. $18 \times 15 =$ _____ | 22. $44 \times 25 =$ _____ |
| 3. $3600 \div 9 =$ _____ | 13. $2400 \div 16 =$ _____ | 23. $864 \div 12 =$ _____ |
| 4. $45 \times 12 =$ _____ | 14. $27 \times 40 =$ _____ | 24. $125 \times 8 =$ _____ |
| 5. $720 \div 8 =$ _____ | 15. $546 \div 7 =$ _____ | 25. $756 \div 7 =$ _____ |
| 6. $16 \times 25 =$ _____ | 16. $23 \times 30 =$ _____ | 26. $48 \times 25 =$ _____ |
| 7. $1000 \div 8 =$ _____ | 17. $1250 \div 25 =$ _____ | 27. $630 \div 15 =$ _____ |
| 8. $60 \times 14 =$ _____ | 18. $19 \times 21 =$ _____ | 28. $26 \times 40 =$ _____ |
| 9. $812 \div 4 =$ _____ | 19. $4000 \div 25 =$ _____ | 29. $1440 \div 12 =$ _____ |
| 10. $35 \times 11 =$ _____ | 20. $32 \times 15 =$ _____ | 30. $55 \times 12 =$ _____ |

Part 4 • Use it**Word help — which operation does the sentence want?**

each / every — usually multiply. 32 boxes with 124 in **each** is 32×124 .

in rows of • shared equally between — divide. **how many are left over** — the remainder.

Write the calculation first, then the answer.

- 1. The hall.** There are **1224 chairs** in the hall, set out in rows of **24**. How many rows are there?

- 2. The books.** Mr Bowen orders **32 boxes** of books with **124 books** in each box. How many books is that altogether?

3. **The pencils.** Mr Bowen has **1000 pencils** and shares them equally between **24 classes**. How many does each class get, and how many are left over?

4. **The order.** The school orders **18 packs** of pens with **236 pens** in each pack. The pens are then shared equally between **24 classrooms**. How many pens does each classroom get?

5. **The tiles.** Mr Bowen tiles a floor with **47 rows** of **26 tiles**. The tiles come in boxes of **20**. How many tiles does he need, and how many boxes must he buy?

6. **The stickers.** Mr Bowen has **4212 stickers** and sticks **18** of them on every page of a sticker book. Each book has **50 pages**. How many pages does he fill, and how many **whole books** is that?